We are using open articles published on the **Arxiv** page that relate to **Generative AI** (**GenAI**) and how it can impact human labor markets and economics. We pass the URLs to be used as documents for our chatbot and load these files into our volume.

```
# we are getting our documents, you could directly upload it to the volumes using UI
pdfs = {'2203.02155.pdf':'https://arxiv.org/pdf/2203.02155.pdf',
        '2302.09419.pdf': 'https://arxiv.org/pdf/2302.09419.pdf',
        '2303.10130.pdf':'https://arxiv.org/pdf/2303.10130.pdf',
        '2302.06476.pdf':'https://arxiv.org/pdf/2312.00506.pdf',
        '2302.06476.pdf':'https://arxiv.org/pdf/2302.06476.pdf',
        '2303.10130.pdf':'https://arxiv.org/pdf/2303.10130.pdf',
        '2209.07753.pdf':'https://arxiv.org/pdf/2209.07753.pdf',
        '2304.07683.pdf':'https://arxiv.org/pdf/2304.07683.pdf'}

for pdf in pdfs.keys():
    load_file(pdfs[pdf], pdf, library_folder)
```

Figure 2.28 – Download PDF files and save them to our volume

Now that we have the documents downloaded and, they are ready to be processed for our chatbot. With that completed, we can move on to our final project: **Multilabel Image Classification**.

Project: Multilabel Image Classification

The MIC project ingests images into Delta tables to fine-tune a pre-trained model from the *Restnet* family to improve its accuracy. We will programmatically download the images from Kaggle and save the data to a volume. To follow along in your workspace, please open the `CH2-01-Downloading Images` notebook:

```
%sh
cd /local_disk0/
kaggle datasets download -d puneet6060/intel-image-classification
```

Figure 2.29 – Downloading data from Kaggle using Databricks magic commands

Now we create the volume folder and unzip the images for our classification project into our volumes. It will take around one hour (as it contains 80K images!) to extract the images from ZIP to `Volumes`.

```
!mkdir {volume_file_path}/intel_image_clf/
!mkdir {volume_file_path}/intel_image_clf/raw_images

# this can take up to a few hours
# or load a few examples to UC on your own if time is a constraint
!unzip -n /local_disk0/intel-image-classification.zip -d {volume_file_path}/intel_image_clf/raw_images
```

Figure 2.30 – Unzipping images into the volumes for this project

We have downloaded or generated all four datasets, and they are ready to be brought into our Bronze layer in the next chapter.